PHYSICS ONLY

17 ENERGY RESOURCES AND ELECTRICITY GENERATION

In this chapter, you will learn about different resources of energy that are available to us on Earth. You will learn that some resources are renewable, while others cannot be replaced once used. As the demand for energy increases with the human population, there is a danger that non-renewable resources will run out. We therefore need to use fuel efficiently and to exploit more renewable resources.



▲ Figure 17.1 Humans consume vast amounts of fuel for transport, as well as heating and cooking.

LEARNING OBJECTIVES

- Describe the energy transfers involved in generating electricity using:
  - wind
  - water
  - geothermal resources
  - solar heating systems
  - solar cells
  - fossil fuels
  - nuclear power
- Describe the advantages and disadvantages of methods of large-scale electricity production from various renewable and non-renewable resources

The demand for energy increases all the time. The growth of the world population means more people need food and warmth. More people want to be able to travel. The fuels that we use to produce energy are being used up too quickly. We must use our remaining fuel supplies efficiently and look for new resources of energy. In Chapter 15, we saw how we can use energy more efficiently in heating our homes. We can make the energy supplies we have available last longer by being energy efficient.

We also need to understand what energy resources we have available on the Earth. In this chapter we shall look at different types of energy resource and the advantages and disadvantages of each resource for generating electricity. In particular we shall distinguish between **renewable** and **non-renewable** energy resources. New resources of energy are being researched all the time to meet our growing needs. We must also consider the effect that the use of different energy resources has on our environment. Some types of energy resource can cause long-term damage to our environment.

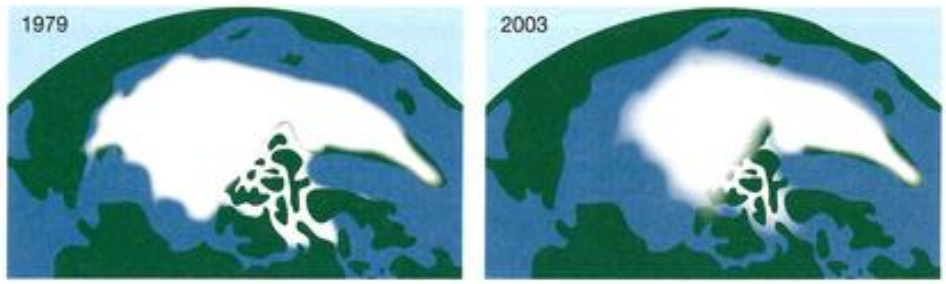
NON-RENEWABLE ENERGY RESOURCES

FOSSIL FUELS



▲ Figure 17.2 Fossil fuels include coal, oil and natural gas.

One of the main energy resources available on our planet is its supply of **fossil fuels**. Coal, oil and natural gas are all fossil fuels. They have been formed in the ground from dead vegetation or tiny creatures by a process that has taken millions of years. Once we have used them, it will take millions of years for new reserves of these fuels to be formed. Fossil fuels are, therefore, examples of non-renewable energy resources.

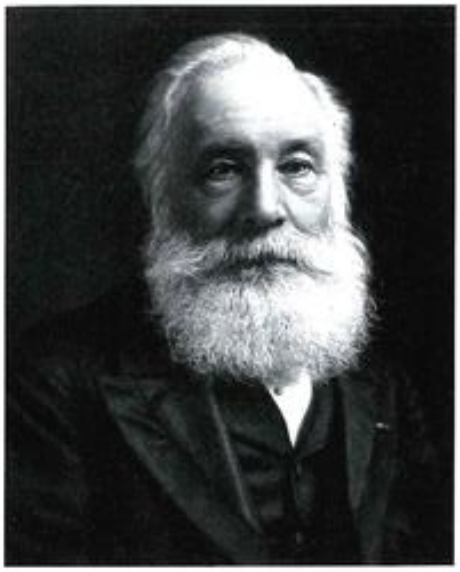


▲ Figure 17.3 Polar ice caps melting

A non-renewable energy resource is one that effectively cannot be replaced once it has been used.

Burning fossil fuels affects the environment, mainly by releasing carbon dioxide into the atmosphere. Carbon dioxide is a greenhouse gas. Greenhouse gases trap the Sun's heat in the Earth's atmosphere and cause the average temperature of the atmosphere to rise. This effect is called global warming and causes changes in the world's climate and melting of the polar ice caps. Burning coal releases more carbon dioxide into the atmosphere than burning oil or gas. Of the fossil fuels, natural gas produces the least carbon dioxide for the same energy output. There is no practical way of avoiding the release of carbon dioxide into the atmosphere when fossil fuels are burned, although some energy companies are researching ways of capturing it and storing it underground.

Most types of coal and oil contain some sulphur. When they are burned, this is converted to sulphur dioxide. Sulphur dioxide is then released into the atmosphere where it combines with water to form acid rain. Acid rain causes damage to people, plants and buildings. It is possible to remove the sulphur from these fuels but this increases the cost of the energy produced. It is also possible to remove sulphur dioxide from the waste gases when the fuel is burned, but this also increases the cost. International agreements are forcing



▲ Figure 17.4 William Perkin discovered the first synthetic dye in 1856 using substances produced from coal tar.

companies that emit large quantities of sulphur dioxide to clean up their waste gases, and acid rain is now less of a problem in Europe than it used to be. Acid rain is still a problem in many countries in the developing world.

Fossil fuels also provide valuable chemicals that can be used in the manufacture of a wide range of useful products (Figure 17.5). Once burned for energy production, these chemical resources are lost permanently. Burning such resources may be a very wasteful way of using them.



▲ Figure 17.5 Many products are manufactured using extracts from oil and coal.

NUCLEAR FUEL

Nuclear reactors use uranium to produce energy. For the nuclear process, a particular form or isotope (see page 223) of uranium is needed. Although a reactor only needs a small amount of uranium fuel, uranium is in limited supply. The uranium in the Earth was formed before the Solar System was formed, so once it has been used there will be no further supplies. It is, therefore, another example of a non-renewable resource.

Power generated from nuclear processes has the advantage of being 'clean'. It is clean because the process does not involve the production of greenhouse or other polluting gases. The cost per unit of electricity is very low, but nuclear power stations are expensive to build. The disadvantages of nuclear power are the risk of accidents and the problem of disposal of **radioactive** material once a power station is finished with it. Accidents that release radioactive materials like uranium and plutonium into the atmosphere pose long-lasting risks to living things.

EXTENSION WORK

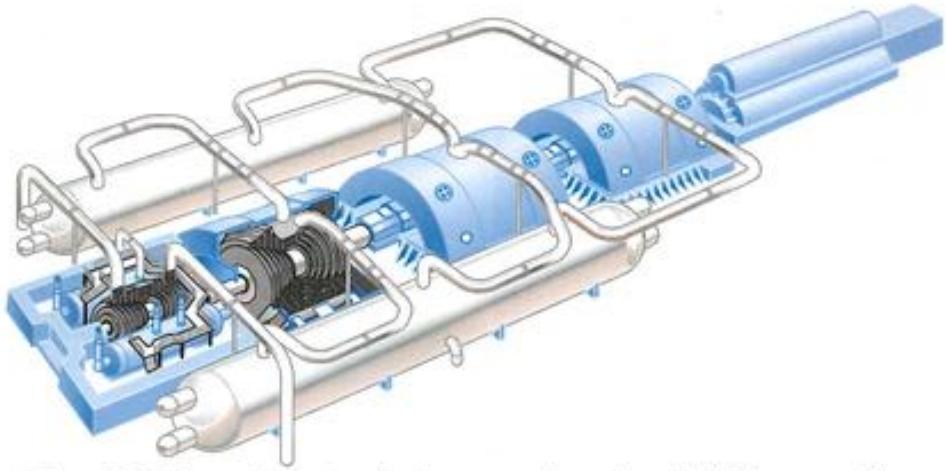
'Fast breeder reactors' are so called because they create more nuclear fuel than they consume, but they still require uranium for their operation. The fuel produced is plutonium. This is extremely dangerous to life and is also a material used for the manufacture of nuclear weapons.

ELECTRICITY

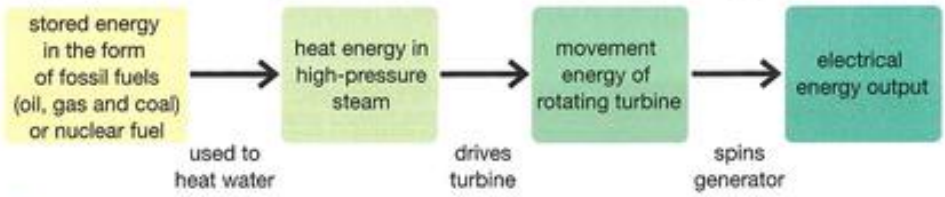
Electricity is not an energy resource, because it has to be generated using other resources of energy. At present, most of the electricity used in the world is generated in power stations like the one shown in Figure 17.6.

Heat from nuclear fuel or from burning fossil fuels is used to heat water. This produces high-pressure steam that makes the blades of a turbine spin. A turbine is like a windmill or a fan, but with many more blades. The turbine is used to turn the generator, which generates the electricity (you will learn more about this in Chapter 21). The energy changes involved are shown in Figure 17.7.

Electricity can also be generated using renewable energy resources.



▲ Figure 17.6 Turbines and generators like these are used to produce electricity in power stations.



▲ Figure 17.7 Several energy changes are involved in producing electricity.

RENEWABLE ENERGY RESOURCES

A renewable energy resource is one that will not run out.

Wood is an example of a renewable energy resource. As wood is cut for fuel, new fast-growing trees are planted to replace those cut down. With careful management the supply of wood fuel can be maintained indefinitely. However, burning wood produces pollution and greenhouse gases. Wood is also more valuable if it is not burned, because it can be used in building or making furniture.

The demand for fuel and our worries about global warming and pollution have made us search for alternative energy resources. We need renewable energy resources that do not pollute the world or contribute to global warming.

HYDROELECTRIC POWER



▲ Figure 17.8 Moving water is a renewable energy resource.

The kinetic energy available in large quantities of moving water has been harnessed (used) for many hundreds of years. Water wheels have been used to transfer the energy stored by water in rivers to grind (break down) corn and power industrial machinery. A different kind of water wheel, called a turbine, is used to turn the generators in a **hydroelectric power** station. These power stations use the stored gravitational potential energy (GPE) of water in high reservoirs built in mountains. The GPE is transferred to kinetic energy (KE) as the water flows down the mountain to the power station below.

The energy transferred in this way is renewable. The Sun causes water to evaporate continuously and to be drawn up into the atmosphere. This water then falls as rain to be collected in reservoirs and used again. Moving water is a renewable resource.

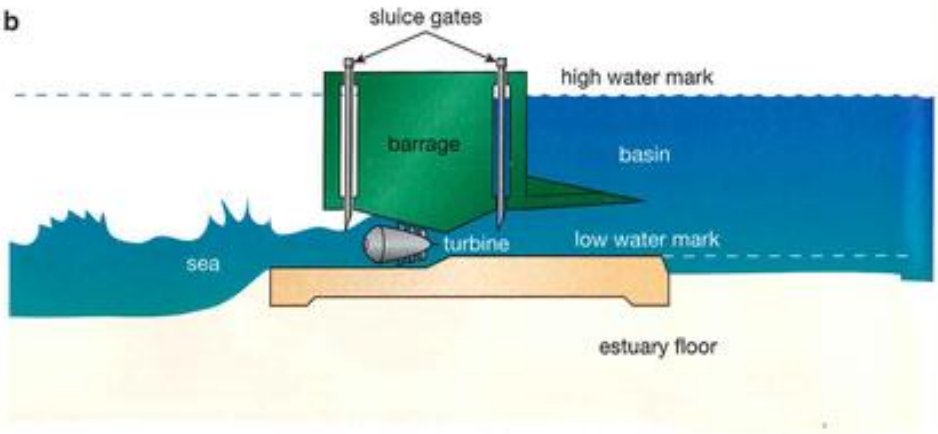
Although hydroelectricity is a very clean, renewable resource, building reservoirs and power stations can spoil the landscape. The reservoir may also destroy or alter the natural habitat for wildlife.

TIDAL POWER

a



b



▲ Figure 17.9 **a** Tidal power station at La Rance, Brittany. **b** Tidal power also involves harnessing (using) the energy in moving water.



▲ Figure 17.10 The turbine has angled blades. As water is forced between the blades, they start to turn. The rotating turbine is connected to a generator to make it turn too.

The tides also involve the movement of huge amounts of water. Tidal power generation schemes, like that at La Rance in Brittany, generate power by turning turbines as the tide flows into a dammed (blocked) river estuary – where a river joins the sea. As the tide falls and the water flows out of the estuary the turbines turn again.

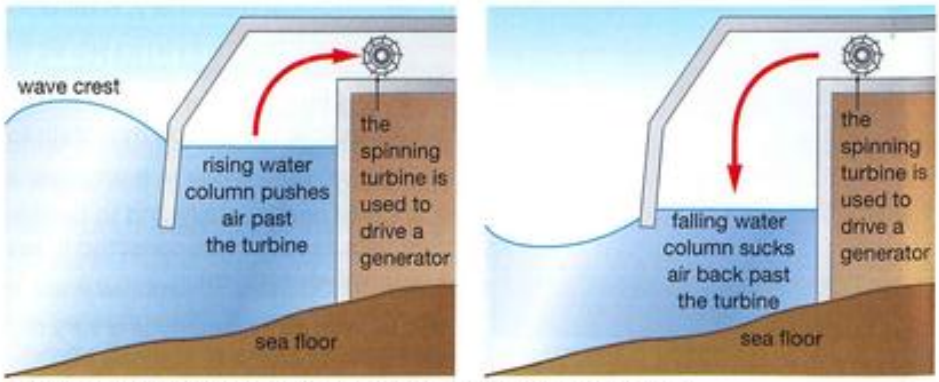
The energy for the movement of the tides is provided by the gravitational pull of the Moon and Sun. This is renewable energy using a small fraction of the continuous supply of gravitational energy.

There are not many places around the world suitable for building dams for tidal energy. If a dam is built, it affects the rise and fall of water in the estuary, and this is likely to damage habitats for wildlife. Some energy companies are now developing 'tidal stream turbines', which are like underwater wind turbines. These will be driven by tidal currents.

WAVE ENERGY

Energy can also be extracted from waves. The continuous movement of the surface of the seas and oceans is the result of a combination of tides and wind. A variety of methods have been developed to make use of the rise and fall of water due to waves. Figure 17.11 shows a system that has been developed to use the energy of waves. Again, this energy is renewable, as the movement energy of the waves is continuously available.

Water power is clean, producing no greenhouse gases or unwanted waste products.



▲ Figure 17.11 An oscillating water column system for using wave energy

WIND POWER



▲ Figure 17.12 Old windmills were a way of using wind energy.

Winds are powered by the Sun's heat energy. Wind is a renewable source of energy that has been used for many centuries. Windmills have been used to grind corn and power machinery like pumps to drain (remove liquid from) lowland areas. Today, wind turbines drive generators to provide electrical energy.

The energy produced is clean, but wind power can only be harvested in regions where the wind blows with enough energy for a significant proportion of the year. Wind farms can cause environmental damage, as they change the appearance of the landscape. They also cause some noise pollution, and may kill birds and bats.

SOLAR POWER PRODUCING ELECTRICITY

Photovoltaic cells (solar cells) transfer light energy directly to electrical energy.

Photovoltaic (PV) cells are around 15% efficient. This means that 15% of the Sun's energy is transferred to useful electric energy. Improvements in PV efficiency and the fact that they are becoming cheaper to produce mean that more PV energy farms are being set up to provide large amounts of electrical energy. PV cells are also used to provide small amounts of electricity for use in places that cannot easily be connected to the electricity mains supply.

Figure 17.13 shows PV cells being used to power the Hubble space telescope, and a PV energy farm.

Advantage: They provide a renewable energy resource that does not produce greenhouse gases.

Disadvantage: Some people think that thousands of square metres of PV cells spoil the countryside. Others are worried that good farming land is lost when PV farms are built.

a



b



▲ Figure 17.13 Photovoltaic cells **a** on the Hubble space telescope, and **b** on an energy farm produce electricity when sunlight falls on them.

SOLAR POWER PRODUCING HEAT

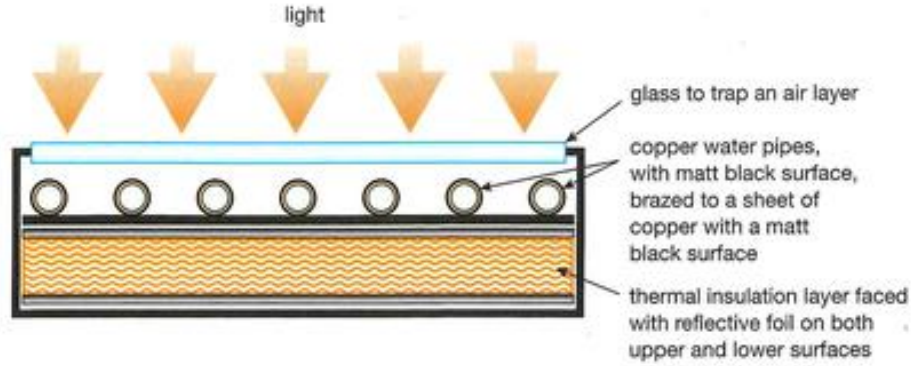


▲ Figure 17.14 Solar heating panels used to heat water

Figure 17.14 shows panels that use heat from the Sun to warm water. They provide small quantities of hot water and reduce the amount of energy that might otherwise come from using non-renewable resources.

The design of efficient solar heating panels involves all the methods of heat transfer discussed in Chapter 15. Solar panels are often used in questions about heat transfer.

Solar heating panels absorb thermal radiation and use it to heat water. The panels are placed to receive the maximum amount of the Sun's energy. In the northern **hemisphere**, they must face south and be angled so that light falls on them as directly as possible for as long as possible. The structure of a typical solar heating panel is shown in Figure 17.15.



▲ Figure 17.15 Here is one type of solar heating panel. They are designed to transfer as much energy to the water passing through them as possible.

EXTENSION WORK

Solar panels used to heat water (and therefore reduce energy used to heat water) are often used as an example of how a knowledge of conduction, convection and radiation can help to make these panels as efficient as possible. The use to generate electricity is described in the solar power boiler shown in Figure 17.16. Other methods are used to concentrate the heating effect of the Sun rays too.

Water is pumped through copper pipes brazed onto a copper sheet. Copper is used because it is an excellent thermal conductor. The surfaces of the sheet and the pipes have a matt (not shiny) black finish as this is the best absorber of heat radiation. The glass traps a layer of air above the copper to help insulate the unit and retain the heat. The backing is also designed to stop heat escaping to the surroundings. This kind of panel is reasonably efficient and the energy produced is more cost efficient than that from photovoltaic cells. Solar heating panels are used widely to provide water heating.



▲ Figure 17.16 A large number of mirrors focus heat from the Sun on a boiler in the tower at the centre. The steam produced drives a turbine to generate electricity.

Electricity can also be generated using solar heating. Curved mirrors are used to focus thermal radiation onto a boiler or pipes containing water to produce steam. The mirrors are controlled to reflect the Sun's heat onto the central tower throughout the day. The steam can be used to drive turbines which can be used to drive electricity generators.

GEOTHERMAL ENERGY

Geothermal energy is thermal energy stored deep inside the Earth. The heat in regions of volcanic activity was produced by the decay of radioactive **elements** like uranium. Volcanoes are evidence of the enormous heat and energy beneath the Earth's surface but do not provide a safe or reliable energy resource. However, heat from the ground can be used safely. In some areas of the world, like Iceland (Figure 17.17), geothermally heated water is readily available in springs and geysers. This is used to drive the turbines in electricity generation stations. The hot water is also used to provide domestic heating by sending it directly to houses. This resource is renewable, does not produce pollution and does not have a great impact on the environment. There are many areas of the world where geothermally heated springs can, and are, being used to provide energy.



▲ Figure 17.17 A geothermal electricity generating station in Iceland, with a huge swimming lagoon heated geothermally

SUPPLY AND DEMAND

Many different energy resources can be used to generate electricity, and they all have advantages and disadvantages. Some of the environmental disadvantages have already been discussed, but supply and demand also need to be taken into account. Different types of power station differ in the speed with which they can meet changes in the demand for electricity.

The demand for electricity varies from hour to hour, day to day and season to season. The way that the demand varies can be predicted to some extent. For example, there is a rise in demand in the early morning as people wake up, turn on lights and heaters and start to make breakfast, and of course in winter the demand for heat is much greater than in summer. Some sudden rises in demand are less predictable. A popular TV programme with an exciting episode can keep millions of viewers in front of their televisions – if they all decide to make a cup of tea as soon as the adverts come on, electricity usage will suddenly increase (electric kettles use a lot of power). The companies that supply electricity must be able to cope with these changes in demand, otherwise they are forced to cut off electricity to some consumers. This is not good for customer relations and, of course, means a reduction in the amount of electricity sold.

Nuclear power stations cannot be turned on instantly. The process of starting the **fission** reaction and heating up the core of the nuclear reactor is a long one. Clearly nuclear power stations cannot meet sudden variations in demand.

Power stations that burn fossil fuels can be started more quickly but can still take many hours to start producing electricity. Coal-fired stations take longer than oil-fired stations to develop the heat required to drive steam through the turbines. Gas-fired stations can respond most quickly to rises in demand.

Hydroelectric power stations provide a very reliable energy resource with the advantage of being able to respond very quickly to changes in the national demand for electricity. Unlike other types of power station, they are able to operate in reverse. This means that they can use extra electricity produced by other power stations that cannot be shut down quickly to pump water back up into the high-level reservoirs. This transfers the electrical energy back to gravitational potential energy, which can then be re-transferred when needed at a later time. This is the only realistic way of 'storing' large amounts of extra electrical energy.

EXTENSION WORK

The US Department of Energy has proposed the building of a nuclear waste storage facility beneath the Yucca Mountains. High-level nuclear waste will be placed in concrete containers that are supposed to hold the nuclear radiation safely for 300 to 1000 years.

Wind power is dependent on the strength, direction and frequency of wind. Although wind farms are located in windy areas, they cannot be relied upon to produce electricity at the times when it is needed most. Tidal power is not available continuously, but the times at which it will be available are predictable.

COST

Planners must also look at the financial costs of electricity generation. Nuclear power uses a relatively cheap fuel. Uranium produces huge amounts of energy, so the cost of energy per unit of fuel used is low. However, building a nuclear power station is very expensive. Nuclear power requires complex technology and very high standards of safety. On top of these 'start-up' costs, planners must also consider the expense of decommissioning (closing down) a nuclear power station at the end of its useful working life. For conventional power stations, this is a routine demolition job, but for nuclear power stations the task is not as straightforward. Radioactive materials must be handled with great care and stored in a way to ensure that none escape.

We see that, although the running costs of nuclear power are relatively low, the pay-back time is very long. (The pay-back time is how long it takes for the income from selling electricity to cover the cost of building the power station.)

The cost of setting up a wind farm is much lower than the cost of building a nuclear power station and there are no fuel costs. However, the amount of energy produced by wind farms is comparatively low. The pay-back time for wind generators is therefore quite long.

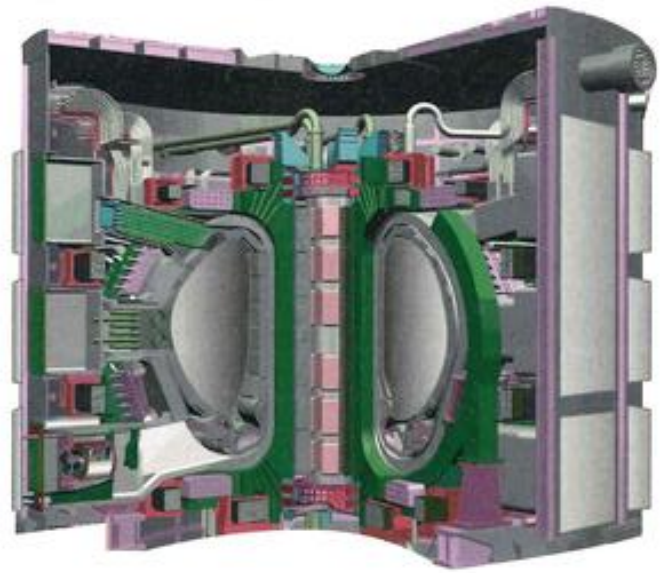
LOOKING AHEAD

Sometimes physics is described as the study of matter and energy. As we have stated in this unit, energy, work and power affect us in everyday life. We use energy to move things around, keep us warm and to manufacture things. As the population of the world increases the demand for energy also increases. Fossil fuels are a non-renewable resource and, although new reserves may be discovered and new methods of extracting these reserves found, they will run out eventually.

Energy is needed to move an object against a force. You will learn more about the forces that govern the way the Universe works: gravity, electromagnetic and the two types of nuclear force. By understanding how to break the forces that hold the atom together, physicists have made power from nuclear fission. Understanding how to force atoms close enough together for them to form more massive atoms will provide us with nuclear power from fusion reactors.

Building experimental fusion reactors is a huge challenge for engineers and scientists. If successful they may provide a long-term solution to the world's demand for energy without the problems of nuclear waste – but a fusion reactor capable of delivering energy on the scale we need has not yet been built.

Scientists continue to search for new solutions to producing energy and to improving the efficiency of the devices we use to cut down on energy waste.



▲ Figure 17.18 Cross-section of a prototype fusion reactor

CHAPTER QUESTIONS

More questions on energy resources can be found at the end of Unit 4 on page 168.

SKILLS CRITICAL THINKING

1 Here is a list of ways of ways of producing electricity:

A wind farms
B coal-fired power stations
C hydroelectric power stations
D nuclear power stations
E gas-fired power stations
F tidal power stations
G using geothermal energy

(5) a State which of these produces 'clean' electricity.

(4) b State which of these uses 'fossil' fuels.

(5) c State which of these uses a renewable energy resource.

d State which of these produces waste that remains dangerous to people and animals for very long periods of time.

SKILLS REASONING

2 From the list in Question 1 choose those methods of producing electricity that are only suitable in certain places and state the type of location required for each method.

SKILLS CRITICAL THINKING

3 Nuclear power generation does not produce greenhouse gases. This is a significant advantage of nuclear power over power stations that burn fossil fuels.

a Describe two other advantages of using nuclear power.

b Describe two disadvantages of using nuclear power.

(6) 4 The demand for electricity varies on an hourly and a daily basis. Sometimes the changes in demand are predictable, but sometimes they are not. Give two examples of predictable changes in demand and two examples of changes in demand that are not predictable.

5 Electricity is generated by a number of different types of power station. Give three examples of different types of power station and compare how well they are able to respond to sudden changes in demand.

END OF PHYSICS ONLY